

2	(a)	<u>Straight line passing through the origin</u> which shows that F and x is proportional	B1
	(b) (i)	Spring constant = $5.6/0.04$ = 140 N m^{-1}	B1
	(b) (ii)	EPE = $\frac{1}{2} (5.6)(0.04)$ = 0.11 J	B1
	(c) (i)	Upthrust = Weight – Tension = $6.2 - 5.6 = 0.60 \text{ N}$	A1
	(c) (ii)	Pressure difference = $0.60/(1.2 \times 10^{-3})$ = 500 Pa	C1 A1
	(d)	If the liquid has a greater density, then <u>the value of upthrust will increase</u> . Since upthrust + spring force = weight, the <u>value of spring force will decrease</u> . Hence the <u>extension will reduce</u> .	M1 A1
	(e)	Percentage uncertainty of h = $\left[\frac{0.4}{6.2} + 2 \left(\frac{0.03}{1.95} \right) + \frac{300}{7800} \right] \times 100\%$ = 13%	C1 A1

3	(a)	The gravitational potential at infinity is set at zero (as there is negligible a	P1
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